

20-June-2026
Sarnali Basak

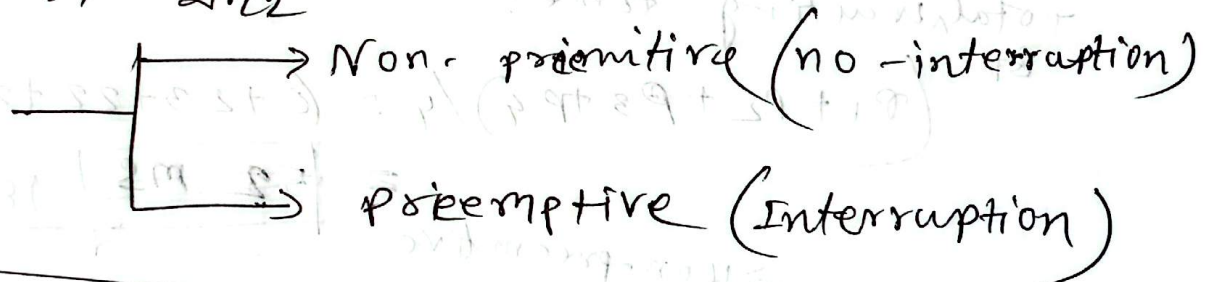
class - 06 (05 class from CT only)

OS - 1st First class

Scheduling Algorithm :

- FCFS Algorithm
- SJF (shortest job-first)
- priority
- Round Robin

FCFS algorithm is a part of

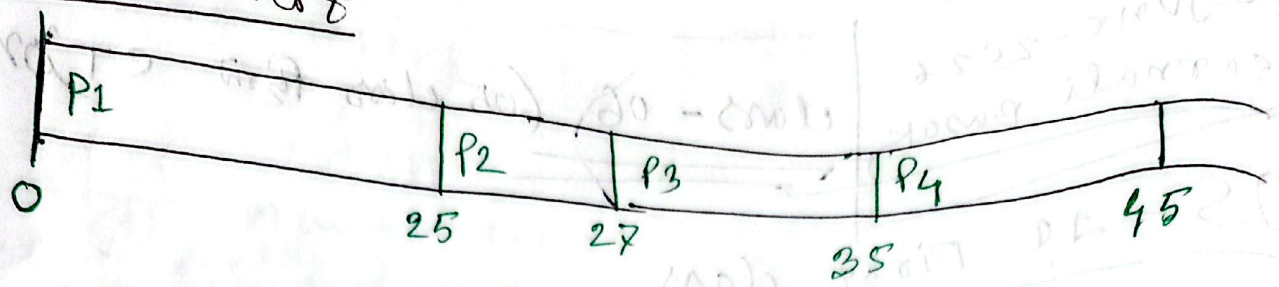


Process ID	Burst Time (ms)	Arrival Time (ms)
P1	25	0
P2	2	2
P3	8	5
P4	10	7

1. FCFS Algorithm (First-come - First-Served)

(P.T.O)

Quantity chart



Average waiting time: Individual process, P1:

$$0 - 0 = 0 \text{ ms}$$

For P2:

$$25 - 2 = 23 \text{ ms}$$

For P3:

$$27 - 5 = 22 \text{ ms}$$

For P4:

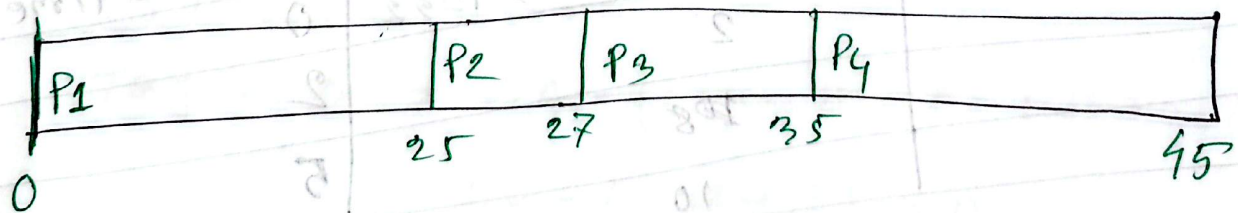
$$35 - 7 = 28 \text{ (ms)}$$

total average waiting time:

$$\begin{aligned} \text{Avg} &= (P_1 + P_2 + P_3 + P_4) / 4 = (0 + 23 + 22 + 28) / 4 \\ &= \boxed{52 \text{ ms}} / 4 = 18.25 \text{ ms} \end{aligned}$$

→ Non-preemptive

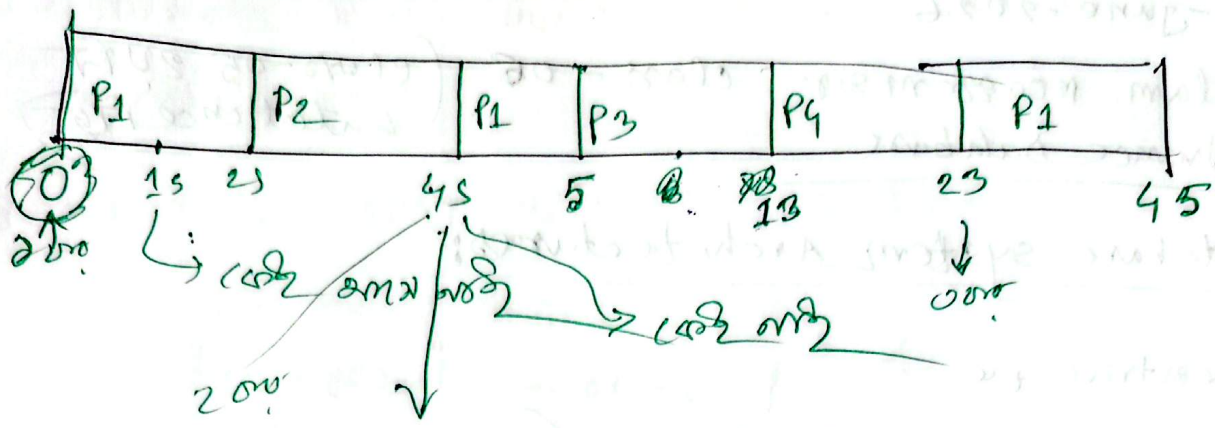
2. SJF Algorithm: shortest burst time or shortest execution time



Average waiting time = 18.25 ms (error not calculation)

preemptive method:

$$P_1: 25 - 2 = 23 - 1 = 22$$



→ $P_1 - 2$ ms

→ $23 - 1/2$ ms $23 - 4$ ms $23 - 4$ ms $23 - 4$ ms

Individual waiting time P_2 :

$$2 - 2 = 0$$

$$P_3: 5 - 5 = 0$$

$$P_4: 13 - 8 = 5 \text{ ms}$$

$$P_1: 0 - 0 = 0$$

$$(4 - 2) = 2 \leftarrow 2 \text{ ms}$$

$$0.5: 23 - 5 = 18$$

$$\text{Total} = 0 + 2 + 18 = 20 \text{ ms}$$

$$\text{AWT} = \frac{0 + 0 + 0 + 6}{4} = \frac{26}{4} = 6.5 \text{ ms}$$